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nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation

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Abstract

Biomedical imaging is a driver of scientific discovery and a core component of medical care and is being stimulated by the field of deep learning. While semantic segmentation algorithms enable image analysis and quantification in many applications, the design of respective specialized solutions is non-trivial and highly dependent on dataset properties and hardware conditions. We developed nnU-Net, a deep learning-based segmentation method that automatically configures itself, including preprocessing, network architecture, training and post-processing for any new task. The key design choices in this process are modeled as a set of fixed parameters, interdependent rules and empirical decisions. Without manual intervention, nnU-Net surpasses most existing approaches, including highly specialized solutions on 23 public datasets used in international biomedical segmentation competitions. We make nnU-Net publicly available as an out-of-the-box tool, rendering state-of-the-art segmentation accessible to a broad audience by requiring neither expert knowledge nor computing resources beyond standard network training.

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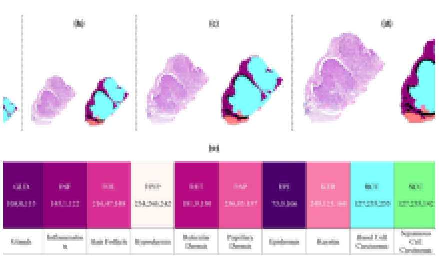

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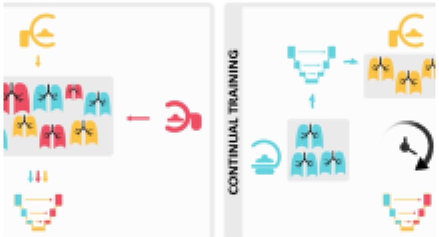

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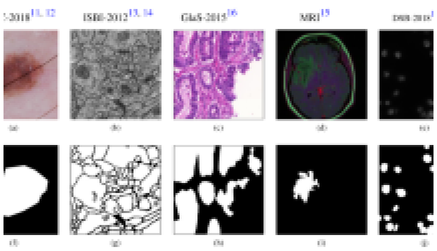

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Data availability

All 23 datasets used in this study are publicly available and can be accessed via their respective challenge websites as follows. D1–D10 Medical Segmentation Decathlon, http://medicaldecathlon.com/; D11 Beyond the Cranial Vault (BCV)-Abdomen, https://www.synapse.org/#!Synapse:syn3193805/wiki/; D12 PROMISE12, https://promise12.grand-challenge.org/; D13 ACDC, https://acdc.creatis.insa-lyon.fr/; D14 LiTS, https://competitions.codalab.org/competitions/17094; D15 MSLes, https://smart-stats-tools.org/lesion-challenge; D16

CHAOS, https://chaos.grand-challenge.org/; D17 KiTS, https://kits19.grand-challenge.org/; D18 SegTHOR, https://competitions.codalab.org/competitions/21145; D19 CREMI, https://cremi.org/; D20–D23 Cell Tracking Challenge, http://celltrackingchallenge.net/.

Code availability

The nnU-Net repository is available as Supplementary Software. Updated versions can be found at https://github.com/mic-dkfz/nnunet. Pretrained models for all datasets used in this study are available for download at https://zenodo.org/record/3734294.

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Contributions

F.I. and P.F.J. conceptualized the method and planned the experiments with the help of S.A.A.K., J.P. and K.H.M.-H. F.I. implemented and configured nnU-Net and conducted the experiments on the 23 selected datasets. F.I. and P.F.J. analyzed the results and performed the KiTS analysis. P.F.J., S.A.A.K. and K.H.M.-H. conceived the communication and presentation of the method. P.F.J. designed and created the figures. P.F.J., F.I. and K.H.M.-H. wrote the paper with contributions from J.P. and S.A.A.K. K.H.M.-H. managed and coordinated the overall project. S.A.A.K. started work on this research as a PhD student at the German Cancer Research Center.

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Competing interests

The authors declare no competing interests.

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